Q1 - 24 June - Shift 1

Let S be the region bounded by the curves $y = x^3$ and $y^2 = x$. The curve $y = 2|x|$ divides S into two regions of areas R_1 and R_2.

If $\max\{R_1, R_2\} = R_2$, then $\frac{R_2}{R_1}$ is equal to _____ .

Space for your notes:

Q2 - 24 June - Shift 2

The area (in sq. units) of the region enclosed between the parabola $y^2 = 2x$ and the line $x + y = 4$ is _____.

Space for your notes:

Q3 - 25 June - Shift 2

The area of the region enclosed between the parabolas $y^2 = 2x - 1$ and $y^2 = 4x - 3$ is

(A) $\frac{1}{3}$ (B) $\frac{1}{6}$

(C) $\frac{2}{3}$ (D) $\frac{3}{4}$

Space for your notes:

Q4 - 26 June - Shift 1

The area bounded by the curve $y = |x^2 - 9|$ and the line $y = 3$ is :

(A) $4\left(2\sqrt{3} + \sqrt{6} - 4\right)$ (B) $4\left(4\sqrt{3} + \sqrt{6} - 4\right)$

(C) $8\left(4\sqrt{3} + 3\sqrt{6} - 9\right)$ (D) $8\left(4\sqrt{3} + \sqrt{6} - 9\right)$

Space for your notes:

Q5 - 26 June - Shift 2

The area of the region bounded by $y^2 = 8x$ and $y^2 = 16(3 - x)$ is equal to :-

(A) $\frac{32}{3}$ (B) $\frac{40}{3}$ (C) 16 (D) 19

Space for your notes:

Q6 - 27 June - Shift 1

Let

$A_1 = \{(x, y) : |x| \le y^2, |x| + 2y \le 8\}$ and

$A_2 = \{(x, y) : |x| + |y| \le k\}$. If 27 (Area A_1) = 5 (Area A_2), then k is equal to :

Space for your notes:

Q7 - 27 June - Shift 2

If the area of the region $\left\{(x, y) : x^{\frac{2}{3}} + y^{\frac{2}{3}} \le 1\ x + y \ge 0,\ y \ge 0\right\}$ is A, then $\frac{256A}{\pi}$ is

Space for your notes:

Q8 - 28 June - Shift 1

The area of the region

$S = \{(x, y) : y^2 \le 8x, y \ge \sqrt{2}x, x \ge 1\}$ is

(A) $\frac{13\sqrt{2}}{6}$ (B) $\frac{11\sqrt{2}}{6}$

(C) $\frac{5\sqrt{2}}{6}$ (D) $\frac{19\sqrt{2}}{6}$

Space for your notes:

Q9 - 28 June - Shift 2

The area of the bounded region enclosed by the curve $y = 3 - \left|x - \frac{1}{2}\right| - |x + 1|$ and the x-axis is

(A) $\frac{9}{4}$ (B) $\frac{45}{16}$

(C) $\frac{27}{8}$ (D) $\frac{63}{16}$

Space for your notes:

Q10 - 29 June - Shift 1

The area enclosed by $y^2 = 8x$ and $y = \sqrt{2}x$ that lies outside the triangle formed by $y = \sqrt{2}x, x = 1, y = 2\sqrt{2}$, is equal to :

(A) $\frac{16\sqrt{2}}{6}$ (B) $\frac{11\sqrt{2}}{6}$

(C) $\frac{13\sqrt{2}}{6}$ (D) $\frac{5\sqrt{2}}{6}$

Space for your notes:

Q11 - 29 June - Shift 2

For real numbers a,b ($a > b > 0$), let

$$Area\left\{(x,y): x^2 + y^2 \le a^2 \text{and } \frac{x^2}{a^2} + \frac{y^2}{b^2} \ge 1\right\} = 30\pi$$

and

$$Area\left\{(x,y): x^2 + y^2 \ge b^2 \text{and } \frac{x^2}{a^2} + \frac{y^2}{b^2} \le 1\right\} = 18\pi$$

Then the value of $(a-b)^2$ is equal to__.

Space for your notes:

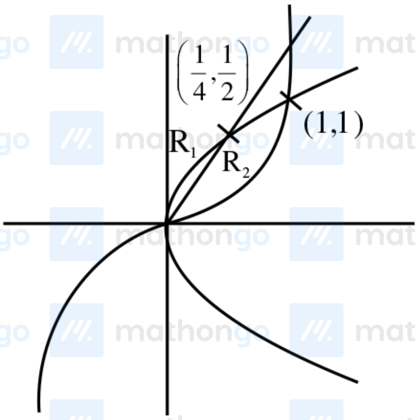
Answer Key

Q1 (19) **Q2** (18) **Q3** (A) **Q4** (D)

Q5 (C) **Q6** (8) **Q7** (36) **Q8** (B)

Q9 (C) **Q10** (C) **Q11** (12)

Q1 (19)



$$S=\int_0^1 \sqrt{x}-x^3$$

$$=\left[\frac{2x^{3/2}}{3}-\frac{x^4}{4}\right]_1^0$$

$$=\frac{5}{12}$$

$$R_1=\int_0^{1/4}(\sqrt{x}-2x)dx$$

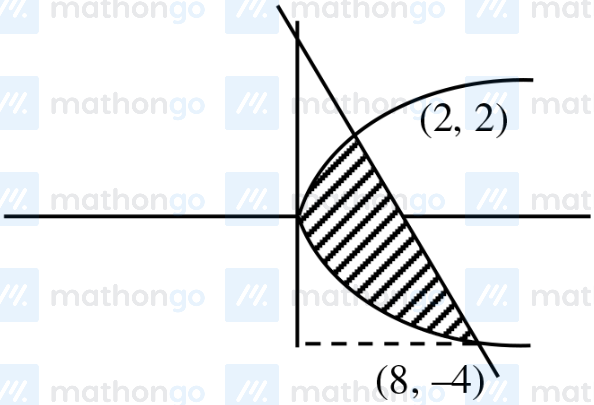
$$=\left[\frac{2x^{3/2}}{3}-x^2\right]_0^{1/4}=\frac{1}{48}$$

$$\therefore\ R_2=\frac{19}{48}$$

$$\text{So, } \frac{R_2}{R_1}=19$$

Q2 (18)

$x = 4 - y$
$y^2 = 2(4 - y)$
$y^2 = 8 - 2y$
$y^2 + 2y - 8 = 0$
$y = -4, y = 2$
$x = 8, x = 2$



$$\int_{-4}^{2}\left[(4-y)-\frac{y^2}{2}\right]dy$$

$$= \left[4y - \frac{y^2}{2} - \frac{y^3}{6}\right]_{-4}^{2}$$

$$= 8 - 2 - \frac{8}{6} + 16 + \frac{16}{2} - \frac{64}{6}$$

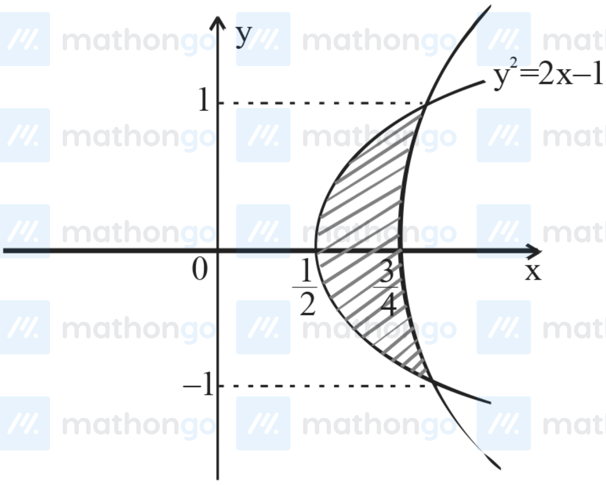
$$= 22 + 8 - \frac{72}{6}$$

$$= 30 - 12 = 18$$

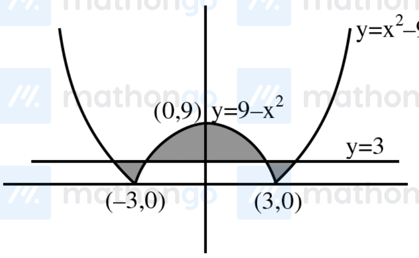
Q3 (A)

Required area $= 2\int_0^1 \left(\frac{y^2+3}{4} - \frac{y^2+1}{2}\right) dy$

$= 2\int_0^1 \frac{1-y^2}{4} dy = \frac{1}{2}\left|y - \frac{y^3}{3}\right|_0^1 = \frac{1}{3}$



Q4 (D)



Area of shaded region

$= 2\int_0^3 \left(\sqrt{9+y} - \sqrt{9-y}\right) dy + 2\int_3^9 \left(\sqrt{9-y}\right) dy$

$= 2\left[\int_0^3 (9+y)^{1/2} dy - \int_0^3 (9-y)^{1/2} dy + \int_3^9 (9-y)^{1/2} dy\right]$

$= 2\left[\frac{2}{3}\left[(9+y)^{3/2}\right]_0^3 + \frac{2}{3}\left[(9-y)^{3/2}\right]_0^3 - \frac{2}{3}\left[(9-y)^{3/2}\right]_3^9\right]$

$= \frac{4}{3}\left[12\sqrt{12} - 27 + 6\sqrt{6} - 27 - \left(0 - 6\sqrt{6}\right)\right]$

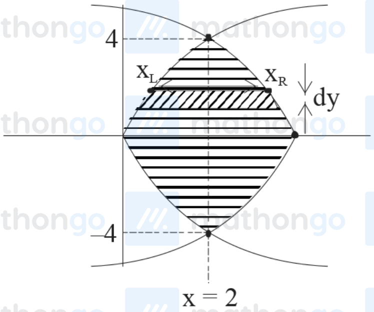
$= \frac{4}{3}\left[24\sqrt{3} + 12\sqrt{6} - 54\right]$

$= 8\left(4\sqrt{3} + 2\sqrt{6} - 9\right)$

Q5 (C)

$y^2 = 8x \; ; \; y^2 = 16(3 - x)$

$y^2 = -16(x - 3)$



finding their intersection pts.

$y^2 = 8x \;\&\; y^2 = -16(x - 3)$

$8x = -16x + 48$

$24x = 48$

$x = 2; \; y = \pm 4$

$$A = 2.\int_0^4 (x_R - x_L)\,dy$$

Required Area

$$= 2.\int_0^4 \left(\underbrace{3 - \frac{y^2}{16}}_{(x_R} - \underbrace{\frac{y^2}{8}}_{x_L)} \right) dy$$

$$= 2\left(3y - \frac{y^3}{3 \times 16} - \frac{y^3}{3 \times 8} \right)_0^4$$

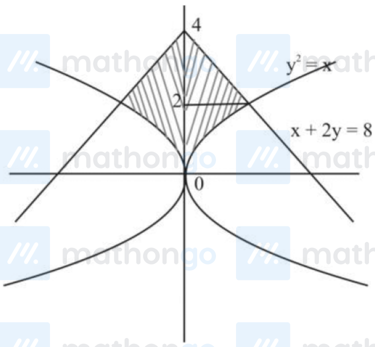
$$= 2\left(3 \times 4 - \frac{4 \times 4 \times 4}{3 \times 16} - \frac{4 \times 4 \times 4 \times 2}{3 \times 8 \times 2} \right)$$

$$= 2\left(12 - \frac{4}{3} - \frac{8}{3} \right) = 2 \times 12\left(1 - \frac{1}{3} \right) \quad = 2 \times 12 \times \frac{2}{3} = 16$$

Q6 (8)

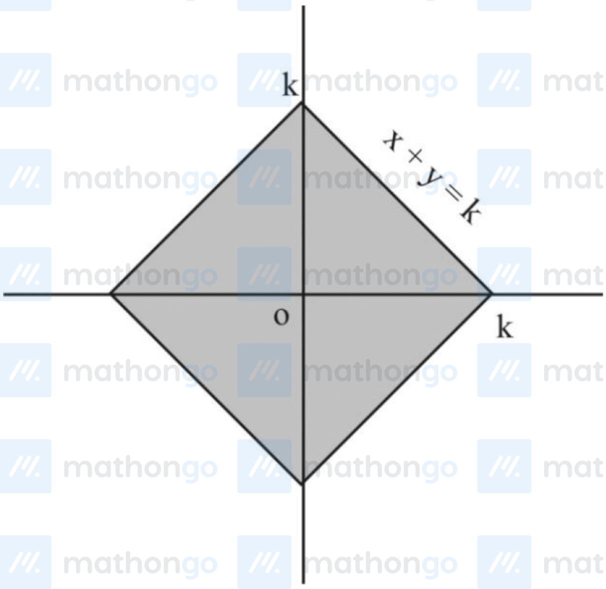
$A_1 = \{(x, y) : |x| \le y^2, |x| + 2y \le 8\}$ and

$A_2 = \{(x, y) : |x| + |y| \le k\}$.



$$area(A_1) = 2\left[\int_0^2 y^2 dy + \int_2^4 (8 - 2y)dy\right]$$

$$= 2\left[\left(\frac{y^3}{3}\right)_0^2 + \left(8y - y^2\right)_2^4\right]$$



$$area(A_1) = 2 \times \frac{20}{3} = \frac{40}{3}$$

Area (A_2) = $4 \times \frac{1}{2} k^2$

Area (A_2) = $2k^2$

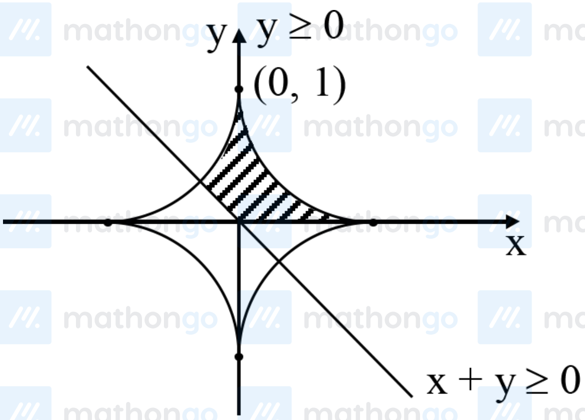
Now

27 (Area A_1) = 5 (Area A_2)

$9 \times 4 = k^2$

k = 6

Q7 (36)



$$A = \frac{3}{2}\int_0^1 (1 - x^{2/3})^{3/2}\,dx$$

Let $x = \sin^3\theta$

$$A = \frac{3}{2}\int_0^{\pi/2} \left(1 - \sin^2\theta\right)^{3/2} .3\sin^2\theta\cos\theta\,d\theta$$

$$= \frac{3}{2}\int_0^{\pi/2} 3\sin^2\theta\cos^4\theta\,d\theta$$

$$= \frac{9}{2}\int_0^{\pi/2} \sin^2\theta\cos^4\theta\,d\theta$$

$$A = \frac{9}{2} \times \frac{1.3.1}{(2+4)(4)(2)} . \frac{\pi}{2}$$

$$\Rightarrow A = \frac{9\pi}{64} \Rightarrow \frac{64A}{\pi} = 9$$

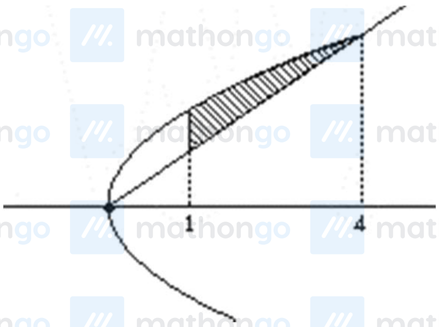
$$\Rightarrow \frac{256A}{\pi} = 36 \textbf{ Ans.}$$

Q8 (B)

$$y^2 = 8x \quad \ldots(1)$$

$$y = \sqrt{2}x \quad \ldots(2)$$

$$y^2 = 2x^2$$



$$\Rightarrow 8x = 2x^2$$
$$\Rightarrow x = 0 \;\&\; 4$$

$$\text{Area} := \int_1^4 2\sqrt{2}\sqrt{x} - \sqrt{2}x \, dx$$

$$= 2\sqrt{2}\left(\frac{x^{\frac{3}{2}}}{3/2}\right)_1^4 - \sqrt{2}\left(\frac{x^2}{2}\right)_1^4$$

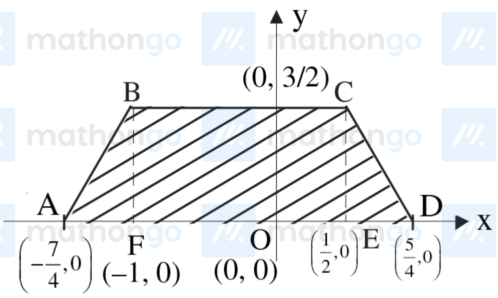
$$= \frac{4\sqrt{2}}{3}(8-1) - \frac{\sqrt{2}}{3}(16-1)$$

$$= \frac{28\sqrt{2}}{3} - \frac{15\sqrt{2}}{2} = \frac{11\sqrt{2}}{6}$$

Q9 (C)

$$y=\begin{cases}3+(x+1)+\left(x-\frac{1}{2}\right), & x<-1\\ 3-(x+1)+\left(x-\frac{1}{2}\right), & -1\le x<\frac{1}{2}\\ 3-(x+1)-\left(x-\frac{1}{2}\right), & \frac{1}{2}\le x\end{cases}$$

$$y=\begin{cases}\frac{7}{2}+2x, & x<-1\\ \frac{3}{2}, & -1\le x<\frac{1}{2}\\ \frac{5}{2}-2x, & \frac{1}{2}\le x\end{cases}$$

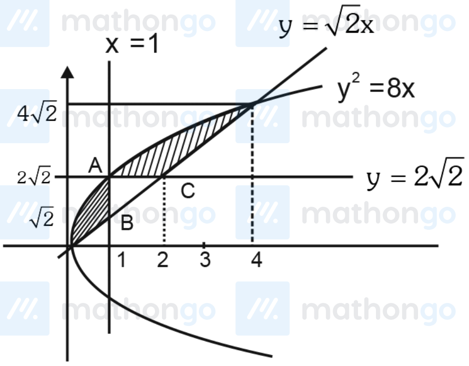


Area bounded = ar ABF + ar BCEF + ar CDE

$$=\frac{1}{2}\left(\frac{3}{4}\right)\left(\frac{3}{2}\right)+\left(\frac{3}{2}\right)\left(\frac{3}{2}\right)+\frac{1}{2}\left(\frac{3}{4}\right)\left(\frac{3}{2}\right)$$

$$=\frac{27}{8}\text{ sq. units.}$$

Q10 (C)



Area of $\Delta ABC = \frac{1}{2}(\sqrt{2})\cdot 1 = \frac{\sqrt{2}}{2}$

So required Area $= \int_0^4 \left(\sqrt{8x} - \sqrt{2}x\right)dx - \frac{\sqrt{2}}{2}$

$= \frac{32\sqrt{2}}{3} - 8\sqrt{2} - \frac{\sqrt{2}}{2} = \frac{13\sqrt{2}}{6}$

Q11 (12)

given $\pi a^2 - \pi ab = 30\pi$ and $\pi ab - \pi b^2 = 18\pi$

on subtracting, we get $(a-b)^2 = a^2 - 2ab + b^2 = 12$